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Module B — Climate association: summary for the manuscript

Scope. Module B (`moduleB_BayPass/`; the cleaned re-organisation of the old `moduleC_*`) tests whether population allele frequencies are associated with the climate covariates **PC1** and **PC2**, at two resolutions: (i) individual SNPs (standard BayPass outlier scan) and (ii) LD-complexity-reduced clusters (eMLGs; one consensus genotype per LD cluster, *sensu* Li et al. 2018). It then calibrates the cluster-level signal against a population-structure null to obtain a valid genome-wide FDR.

Primary configuration throughout: Åland-excluded (19 populations, 154 individuals), fixed LD-pruned Ω covariance (`withOmega`); marker-level significance = **BF(dB) ≥ 15** . Full (20-population) and internally-estimated- Ω runs are sensitivity comparisons. Data: 1,114,340 MAF-filtered SNPs $\rightarrow$ 32,840 LD clusters with an eMLG (≥ 5 loci; `ld_w=0.025`, `cM=0.5`); genome-wide median cluster size = 9 markers.

1. Main-text conclusions (short — headline only)

- Standard single-SNP climate-outlier scans over-call large, differentiated regions.** Under the size-dependent member-count criterion (a cluster is a “candidate” when ≥ 5 member SNPs reach BF ≥ 15), candidate clusters are 7–350 $\times$ larger than the genome-wide median and sit near the background significant-marker rate — i.e. differentiated blocks, not focal climate peaks.
- LD-complexity reduction increases power by collapsing ~ 1.1 M SNP tests to 32,840 cluster tests ($\sim 34\times$ lower multiple-testing burden).** Testing the eMLG (cluster consensus) recovers coherent cluster-level climate signals that the SNP scan cannot: **28 “eMLG-only” clusters** reach eMLG BF ≥ 15 with *no* individually significant member SNP.
- After a population-structure null and a valid genome-wide FDR, the climate signal is largely a structure/differentiation by-product.** Against 10,000 Ω -structured null covariates, **none of the SNP-based candidates survive** (the flagship Chr24 block F63637, BF = 38.4, is beaten by 12/10,000 structure nulls); only **5 PC2 “eMLG-only” clusters** resist structure (set FDR ≈ 0.66 — suggestive, not significant), and **PC1 shows nothing**.
- Net:** no locus reaches a defensible FDR for climate association after structure control. This is a clean negative that both (a) corroborates the weak climate link seen elsewhere and (b) is itself a demonstration of the value of LD-complexity reduction — single-SNP outliers behave as expected false positives, and the only residual signal is the reduced-test, aggregated eMLG signal.

2. Methods (supplementary by default)

2.1 SNP-level BayPass climate association

Association of per-population allele frequencies with PC1/PC2 was evaluated with the standard covariate (IS) model in BayPass v3 (`-nocovscaling -nval 500 -burnin 5000 -thin 25`), evidence reported as Bayes factors on the deciban scale, BF(dB). Eight configurations crossed $\{\text{full, Åland-excluded}\} \times \{\text{PC1, PC2}\} \times \{\Omega \text{ estimated internally from the full set (noOmega), } \Omega \text{ pre-estimated from the LD-pruned set and supplied fixed (withOmega)}\}$. Output rows were matched positionally to the input markers (row-count and strictly-increasing-index asserts). Åland-excluded $\times$ withOmega is primary. Background significant-marker rate (BF ≥ 15): **PC1 0.17 %**, **PC2 0.15 %**. Scripts:

moduleB_BayPass/R/moduleB_prepare_{with_aland,aland_excluded}.R, moduleB_write_baypass_inputs.R, {with_aland,aland_excluded}/run_baypass.sh, moduleB_analyse_baypass.R.

2.2 PC provenance and the ancestry confound

PC1/PC2 are per-population **climate** covariates (within-population SD = 0, i.e. not a per-individual genetic PCA). Rank correlation with per-population *aquilonia* ancestry ($\approx 49k$ diagnostic markers): **PC1** = **-0.229**, **PC2** = **+0.568**. The PC2- ancestry link is driven by **Åland** (+0.568 $\rightarrow$ +0.384 when dropped), not Sielva (+0.568 $\rightarrow$ +0.642 when dropped). Controls: fixed Ω (conditions out shared structure, present in both runs) and the Åland-excluded primary analysis. Associations are therefore interpreted as **conditional** associations, not direct evidence of local adaptation. Script: moduleB_BayPass/R/moduleB_ancestry_confound.R.

2.3 LD-complexity reduction and the eMLG association (Li et al. 2018)

Each ≥ 5 -locus LD cluster was reduced to one **eMLG** (per-individual consensus dosage, rounded to {0,1,2}), aggregated to per-population diploid allele counts, and run through the *same* BayPass covariate model with the *same* fixed Ω , covariates and pool sizes (19-population order asserted identical). This tests whether a cluster tracks climate **as a unit** rather than via a handful of individually significant members. Because Ω is a property of the populations and BayPass co-estimates a genome-wide allele-frequency prior from *all* markers in a run, this must be run on the full 32,840-eMLG set (verified: subsetting to candidates shifts BF by up to tens of dB). Scripts: moduleB_BayPass/R/moduleB_write_eMLG_baypass_inputs.R, run_baypass_eMLG.sh, moduleB_eMLG_manhattan.R. (Note: BayPass eBPis is degenerate here — essentially all eMLGs $p < 0.05$ — so BF(dB), calibrated by the sim-null below, is used, not eBPis.)

2.4 Ω -structured simulation null and genome-wide sim-FDR

Population structure inflates BF for differentiated clusters (structure-only null covariates reach eMLG BF up to 44). To calibrate, **10,000 null “climate” covariates** were drawn from the fixed Ω (structure autocorrelation, no genotype link) and run through the eMLG covariate model in 10 batches of 1,000 on the full set (validated equivalent to a single 10,000-covariate run and to separate runs; marker-level BF depends on the marker set but not the covariate count). For each eMLG we count k = number of the 10,000 structure nulls with BF $\geq$ the observed climate BF (k_1 for PC1, k_2 for PC2). A **floor-survivor** ($k = 0$) is a cluster whose real climate BF beats *all* 10,000 structure nulls. This is a valid permutation-style genome-wide FDR: under the pure-structure null the observed BF is exchangeable with its nulls, so the expected number of floor-survivors is $N/(NSIM+1) = 32,840/10,001 \approx 3.28$ per axis, and the FDR of the observed floor-survivor set = $3.28 / (\text{number observed})$. Any cluster above the floor ($k \geq 1$) cannot survive FDR and is discarded. Script: moduleB_BayPass/R/moduleB_eMLG_null.R $\rightarrow$ moduleB_BayPass/data/moduleB_eMLG_null.rds. (Scoping FDR to a pre-screened subset is invalid — selection on the same statistic — so the genome-wide denominator is used.)

3. Figures (primary analyses)

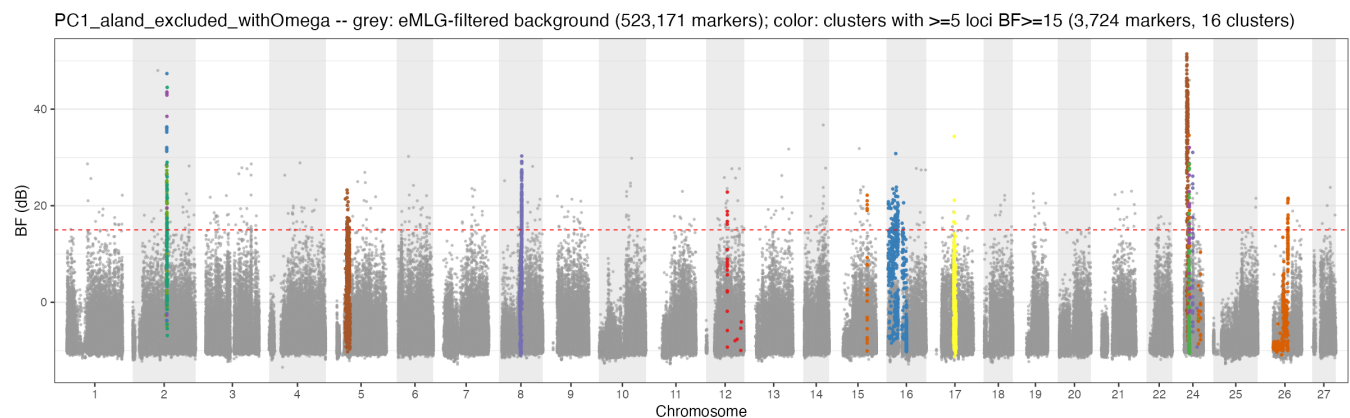


Figure 1: **SNP-level Manhattan, PC1** (primary config). Grey = eMLG-filtered background; coloured = clusters with ≥ 5 member SNPs at BF(dB) ≥ 15 (the member-count candidate set), coloured by LD cluster; dashed line BF = 15.

Compiled candidate tables (standalone PDFs): tables/climate_candidate_clusters.pdf, tables/climate_candidate_full.pdf

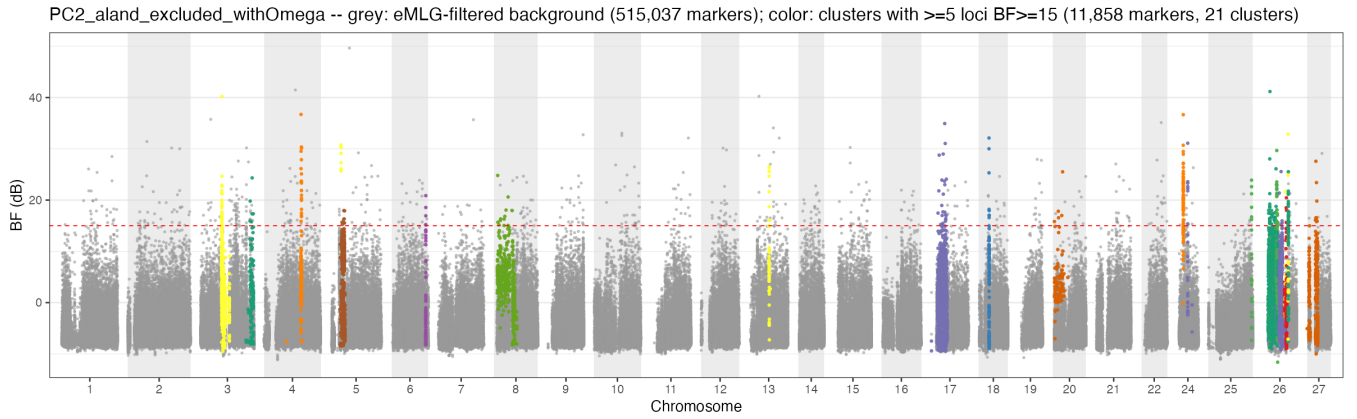


Figure 2: **SNP-level Manhattan, PC2** (primary config). Colouring as above.

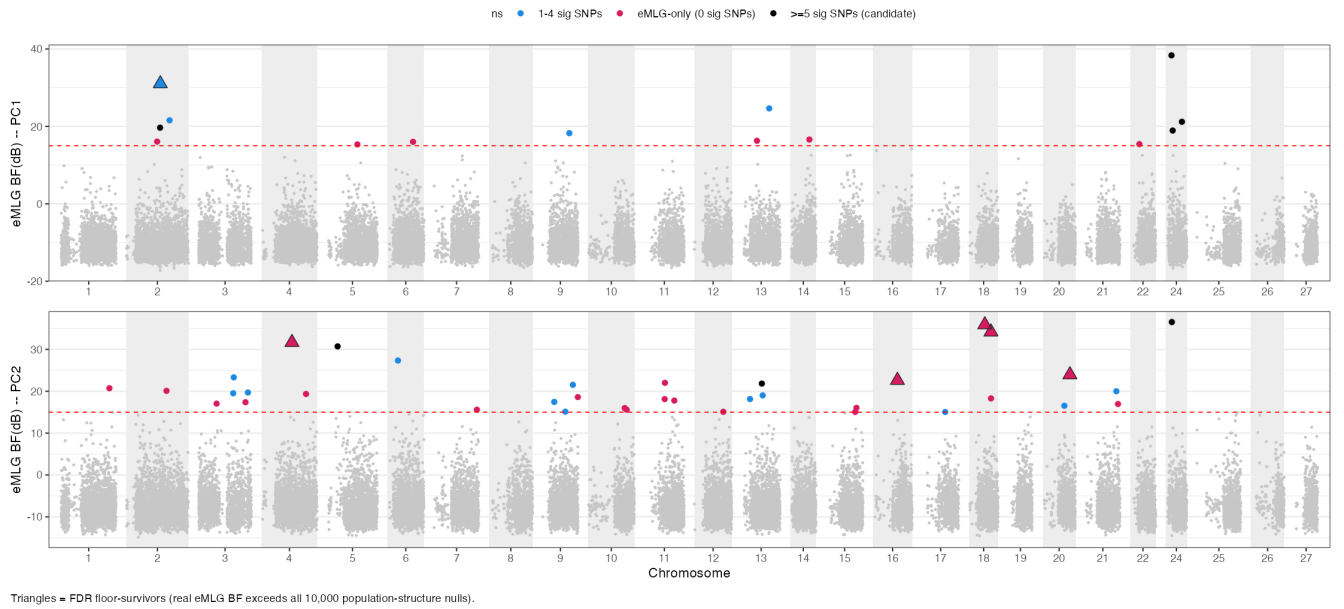


Figure 3: **eMLG-level Manhattan (complexity-reduced)**, primary config. One point per eMLG at its representative position. Red = eMLG BF ≥ 15 with 0 significant member SNPs (eMLG-only); blue = 1-4 sig SNPs; black = ≥ 5 (member-count candidates); grey = ns; dashed line BF = 15. **Larger triangles = the sim-FDR floor-survivors** (real eMLG BF exceeds all 10,000 Ω -structured population-structure nulls): F7388 on PC1; F14310, F49926, F53948, F54421, F58676 on PC2.

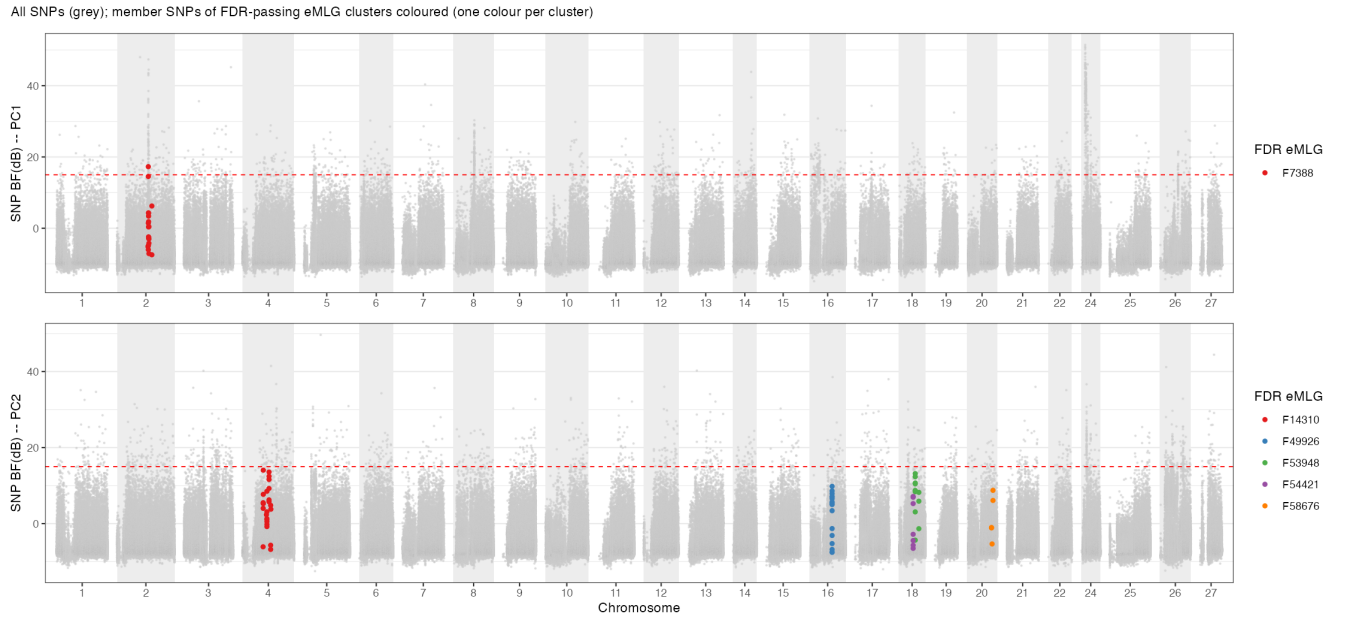


Figure 4: **Final Manhattan — FDR-passing eMLG clusters over all SNPs.** All 1,114,340 SNPs in grey (BF per axis); the member SNPs of the sim-FDR floor-surviving eMLG clusters are coloured, one colour per cluster (PC1: F7388; PC2: F14310, F49926, F53948, F54421, F58676). Most member SNPs lie *below* BF = 15 — the cluster-level signal is not driven by individually significant SNPs. Dashed line BF = 15.

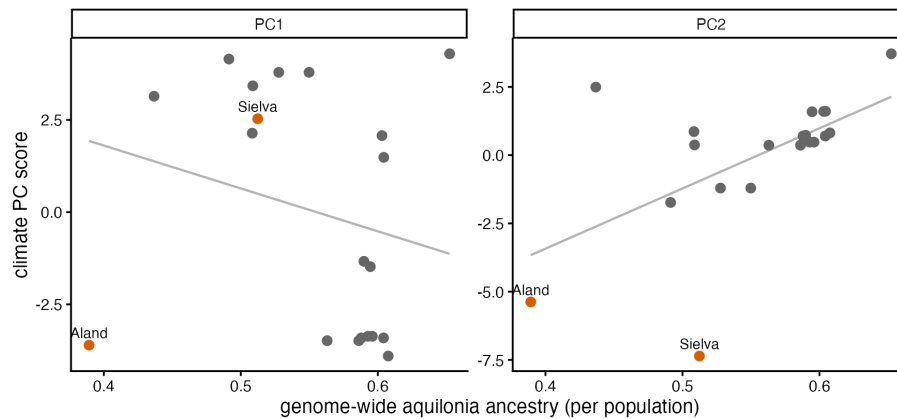


Figure 5: **PC–ancestry confound.** Per-population climate PC vs genome-wide *aquilonia* ancestry; Åland/Sielva highlighted.

4. Tables (complete)

Table S1 — Putative candidate counts across the 8 BayPass configurations

Member-count criterion only (a cluster is a candidate when ≥ 5 , or ≥ 10 , member SNPs reach $BF \geq 15$). Genome-wide median cluster size = 9 markers.

Population set	Ω (covariance)	Axis	Candidates (≥ 5)	Candidates (≥ 10)	Median candidate size
Åland-excluded (19)	Internal (full data)	PC1	14	7	68
Åland-excluded (19)	Internal (full data)	PC2	16	7	116
Åland-excluded (19)	LD-pruned (fixed) [PRIMARY]	PC1	16	10	68
Åland-excluded (19)	LD-pruned (fixed) [PRIMARY]	PC2	21	8	118
Full (20)	Internal (full data)	PC1	9	5	272
Full (20)	Internal (full data)	PC2	43	24	94
Full (20)	LD-pruned (fixed)	PC1	7	5	486
Full (20)	LD-pruned (fixed)	PC2	53	36	89

Table S2 — Member-count candidates, primary config (35 clusters)

Entry = ≥ 5 member SNPs at $BF \geq 15$ on PC1 or PC2. Columns: member SNP count/% on the entry axis; eMLG BF(dB) and β (cluster-consensus association); XtX (among-population differentiation); DI% (proportion of members with DiagnosticIndex > -25); **simnull_k** = # of 10,000 structure nulls $\geq$ observed eMLG BF on the entry axis; **gate15** = eMLG BF ≥ 15 ; **simFDR** = floor-survivor ($k = 0$). Ordered by eMLG BF.

cluster	Chr	pos (Mb)	size	axis	SNP nsig	SNP %	eMLG BF	eMLG β	XtX	DI%	simnull_k	gate15	simFDR
F63637	Chr24	1.3–1.8	272	PC1	205	75.4	38.4	0.094	37	59	12	✓	—
F63772	Chr24	1.8–1.8	83	PC2	60	72.3	36.5	0.122	26	0	1	✓	—
F17762	Chr5	3.0–3.3	11	PC2	7	63.6	30.7	0.123	26	27	1	✓	—
F41962	Chr13	5.8–6.2	50	PC2	7	14.0	21.8	0.095	26	0	8	✓	—
F64355	Chr24	1.9–5.4	35	PC1	7	20.0	21.2	0.051	23	6	9	✓	—
F6554	Chr2	10.0–10.1	24	PC1	12	50.0	19.7	0.067	20	0	9	✓	—
F63812	Chr24	2.0–2.3	23	PC1	10	43.5	18.9	0.057	23	22	18	✓	—
F27646	Chr8	6.2–6.8	486	PC1	108	22.2	14.9	0.058	23	46	119	—	—
F67035	Chr26	1.6–4.7	397	PC2	19	4.8	14.4	0.076	23	73	6	—	—
F49451	Chr16	0.4–5.9	522	PC1	69	13.2	13.7	0.055	21	0	16	—	—
F17747	Chr5	2.8–4.3	690	both	24	3.5	11.5	0.088	23	14	296	—	—
F11460	Chr3	6.2–8.5	1143	PC2	25	2.2	10.5	0.079	21	1	45	—	—
F66997	Chr26	1.4–4.7	3042	PC2	37	1.2	10.1	0.082	23	17	231	—	—
F48626	Chr15	8.2–8.2	19	PC1	5	26.3	8.2	0.050	25	0	401	—	—
F14815	Chr4	6.4–11.0	99	PC2	19	19.2	7.6	0.078	30	28	557	—	—
F27668	Chr8	1.0–6.9	400	PC2	12	3.0	7.2	0.061	24	0	568	—	—
F64102	Chr24	3.1–4.4	28	both	7	25.0	5.7	0.074	26	4	613	—	—
F67238	Chr26	6.6–7.5	42	PC2	7	16.7	5.0	0.062	21	12	453	—	—
F51523	Chr17	0.0–4.8	3183	PC2	24	0.8	4.3	0.058	19	0	561	—	—
F57265	Chr20	0.4–4.4	87	PC2	5	5.7	3.7	0.063	21	0	985	—	—
F40144	Chr12	6.3–10.4	22	PC1	5	22.7	1.6	0.039	20	5	746	—	—
F13363	Chr3	13.4–15.7	131	PC2	6	4.6	0.7	0.048	23	1	2012	—	—
F67071	Chr26	1.5–5.9	379	PC1	13	3.4	−2.5	0.030	23	20	3400	—	—
F51449	Chr17	5.1–5.7	893	PC1	6	0.7	−4.0	0.028	20	5	4243	—	—
F64194	Chr24	1.9–5.1	182	PC1	14	7.7	−6.4	0.024	26	8	6390	—	—

cluster	Chr	pos (Mb)	size	axis	SNP nsig	SNP %	eMLG BF	eMLG β	XtX	DI%	simnull_k	gate15	simFDR
F68272	Chr27	0.1–3.2	496	PC2	8	1.6	−10.9	0.018	19	12	9991	—	—
F24440	Chr6	10.1–10.2	82	PC2	5	6.1	−11.2	0.018	23	33	9989	—	—
F67028	Chr26	1.8–6.7	1236	PC2	5	0.4	−11.9	0.016	22	41	9997	—	—
F67081	Chr26	6.3–7.0	449	PC2	5	1.1	−12.6	0.014	22	51	10000	—	—
F65755	Chr25	12.7–12.7	20	PC2	6	30.0	−13.4	0.012	21	5	10000	—	—
F67249	Chr26	7.4–7.5	71	PC2	26	36.6	−14.1	0.011	22	62	10000	—	—
F54612	Chr18	3.0–3.1	118	PC2	8	6.8	−14.5	0.009	22	42	10000	—	—
F7206	Chr2	9.7–10.1	64	PC1	28	43.8	−15.6	0.007	26	2	10000	—	—
F7494	Chr2	9.9–10.2	71	PC1	22	31.0	−16.4	0.006	23	0	10000	—	—
F6596	Chr2	10.0–10.1	14	PC1	5	35.7	−17.2	0.005	23	0	10000	—	—

Note the demotion pattern: several high member-% clusters (F6596, F7494, F7206, F67249, F65755) have strongly **negative** eMLG BF — a handful of member SNPs cross the threshold but the cluster consensus anti-associates. **None of the 35 survive the sim-FDR** (simFDR all —).

Table S3 — eMLG outliers (BF ≥ 15), primary config (51 clusters)

The complexity-reduced view. **support**: eMLG-only = 0 significant member SNPs; 1–4 SNPs; ≥ 5 SNPs = also a member-count candidate. **simnull_k** = # of 10,000 structure nulls $\geq$ observed; **floor** = k = 0 (survives the structure null).

cluster	Chr	pos (bp)	size	axis	SNP nsig	SNP %	eMLG BF	support	simnull_k	floor
F63637	Chr24	1673178	272	PC1	205	75.4	38.4	≥ 5 SNPs(cand)	12	—
F7388	Chr2	10138607	20	PC1	1	5.0	31.0	1–4 SNPs	0	✓
F43576	Chr13	8123831	6	PC1	2	33.3	24.6	1–4 SNPs	5	—
F7404	Chr2	12899649	28	PC1	2	7.1	21.6	1–4 SNPs	12	—
F64355	Chr24	4784788	35	PC1	7	20.0	21.2	≥ 5 SNPs(cand)	9	—
F6554	Chr2	10056976	24	PC1	12	50.0	19.7	≥ 5 SNPs(cand)	9	—
F63812	Chr24	2032351	23	PC1	10	43.5	18.9	≥ 5 SNPs(cand)	18	—
F31722	Chr9	7966699	5	PC1	2	40.0	18.2	1–4 SNPs	11	—
F45693	Chr14	5672151	10	PC1	0	0.0	16.6	eMLG-only	34	—
F41301	Chr13	4478466	12	PC1	0	0.0	16.3	eMLG-only	48	—
F7080	Chr2	9189106	41	PC1	0	0.0	16.1	eMLG-only	97	—
F23332	Chr6	7613474	8	PC1	0	0.0	16.0	eMLG-only	75	—
F61797	Chr22	2726658	42	PC1	0	0.0	15.4	eMLG-only	64	—
F20510	Chr5	8946084	13	PC1	0	0.0	15.3	eMLG-only	110	—
F63772	Chr24	1777883	83	PC2	60	72.3	36.5	≥ 5 SNPs(cand)	1	—
F54421	Chr18	4661936	7	PC2	0	0.0	35.9	eMLG-only	0	✓
F53948	Chr18	6499523	11	PC2	0	0.0	34.2	eMLG-only	0	✓
F14310	Chr4	9075597	24	PC2	0	0.0	31.7	eMLG-only	0	✓
F17762	Chr5	3040193	11	PC2	7	63.6	30.7	≥ 5 SNPs(cand)	1	—
F21890	Chr6	3096997	5	PC2	2	40.0	27.3	1–4 SNPs	6	—
F58676	Chr20	8129042	5	PC2	0	0.0	23.9	eMLG-only	0	✓
F12448	Chr3	10504842	6	PC2	2	33.3	23.3	1–4 SNPs	8	—
F49926	Chr16	7271131	17	PC2	0	0.0	22.6	eMLG-only	0	✓
F35840	Chr11	6084388	14	PC2	0	0.0	22.0	eMLG-only	1	—

cluster	Chr	pos (bp)	size	axis	SNP nsig	SNP %	eMLG BF	support	simnull_k	floor
F41962	Chr13	5926761	50	PC2	7	14.0	21.8	≥5 SNPs(cand)	8	—
F32802	Chr9	9005754	9	PC2	4	44.4	21.5	1–4 SNPs	14	—
F3838	Chr1	14393778	12	PC2	0	0.0	20.7	eMLG- only	3	—
F6754	Chr2	11986955	10	PC2	0	0.0	20.1	eMLG- only	1	—
F60466	Chr21	9159798	5	PC2	1	20.0	20.0	1–4 SNPs	13	—
F13395	Chr3	14731412	9	PC2	4	44.4	19.7	1–4 SNPs	3	—
F12565	Chr3	10321642	6	PC2	1	16.7	19.5	1–4 SNPs	7	—
F16915	Chr4	13266783	15	PC2	0	0.0	19.3	eMLG- only	1	—
F41903	Chr13	6175093	7	PC2	1	14.3	19.0	1–4 SNPs	34	—
F31253	Chr9	10491866	11	PC2	0	0.0	18.6	eMLG- only	6	—
F53378	Chr18	6514541	5	PC2	0	0.0	18.3	eMLG- only	2	—
F40827	Chr13	2402018	57	PC2	1	1.8	18.1	1–4 SNPs	14	—
F35768	Chr11	6010746	6	PC2	0	0.0	18.1	eMLG- only	13	—
F36504	Chr11	8883438	9	PC2	0	0.0	17.8	eMLG- only	2	—
F30134	Chr9	3458646	10	PC2	1	10.0	17.5	1–4 SNPs	2	—
F13266	Chr3	14003903	16	PC2	0	0.0	17.4	eMLG- only	22	—
F11525	Chr3	5371732	5	PC2	0	0.0	17.0	eMLG- only	2	—
F61514	Chr21	9665372	16	PC2	0	0.0	17.0	eMLG- only	36	—
F57853	Chr20	6463479	6	PC2	1	16.7	16.5	1–4 SNPs	61	—
F48934	Chr15	8997189	6	PC2	0	0.0	16.0	eMLG- only	27	—
F34281	Chr10	10908690	8	PC2	0	0.0	16.0	eMLG- only	33	—
F34785	Chr10	11485945	8	PC2	0	0.0	15.6	eMLG- only	43	—
F26091	Chr7	12741603	5	PC2	0	0.0	15.6	eMLG- only	13	—
F32290	Chr9	6756230	8	PC2	1	12.5	15.1	1–4 SNPs	11	—
F39601	Chr12	8679100	12	PC2	0	0.0	15.1	eMLG- only	16	—
F48676	Chr15	8634509	5	PC2	0	0.0	15.1	eMLG- only	22	—
F52036	Chr17	6835729	8	PC2	2	25.0	15.0	1–4 SNPs	54	—

Composition of the 51 outliers: PC1 = 14 (4 member-count candidates, 4 with 1–4 sig SNPs, 6 eMLG-only); PC2 = 37 (3 candidates, 12 with 1–4 sig SNPs, 22 eMLG-only).

Table S4 — Ω -structured sim-FDR (10,000 nulls), primary config

Floor-survivor = eMLG whose real climate BF exceeds all 10,000 population-structure nulls ($k = 0$). Expected floor count under the pure-structure null = 3.28/axis; set FDR = 3.28 / (observed).

axis	floor-survivors	expected (null)	set FDR	survivors
PC1	1	3.28	$\approx 3.28 (\rightarrow 1)$	F7388
PC2	5	3.28	0.66	F54421, F53948, F14310, F58676, F49926

All five PC2 survivors are **eMLG-only** clusters (no individually significant member SNP), max eMLG BF 35.9 (F54421). None of the 7 SNP-based member-count candidates survive (Table S2, **simnull_k** column: F63637 $k = 12$, F63812 $k = 18$, ...).

5. Supplementary text / interpretation notes

- **Size dependence of the member-count criterion.** Median candidate cluster size (68–486 markers) vastly exceeds the genome-wide median (9), because an *absolute* count of significant members scales with cluster size.

This is why the member-count set is treated as descriptive/putative and the quantitative inference uses the eMLG.

- **Why the sim-null, and why it must be genome-wide.** The eMLG set is enriched for strong-LD / differentiated blocks, exactly where climate PCs (partly confounded with ancestry, §2.2) leak through Ω . Structure-only null covariates reach eMLG BF up to 44, so $\text{BF} \geq 15$ alone is not self-sufficient. FDR must use the full 32,840-eMLG denominator; scoping it to the pre-screened outliers is selection on the test statistic and is invalid (equivalently, it is the same error as subsetting the BayPass input, which also changes the co-estimated allele-frequency prior and the BF).
- **Resolution.** At 1,000 sims the empirical-p floor ($1/1001$) equals the null expectation ($32.8 \approx \text{observed } 32$) $\rightarrow$ $\text{FDR} \approx 1$ (nothing). 10,000 sims lower the null expectation to 3.28, so **set** $\text{FDR} = 3.28/m$ and $q < 1$ becomes attainable ($\text{PC2} = 0.66$). Reaching $q < 0.05$ would require ≥ 66 floor-survivors, which the data do not deliver.
- **Interpretive framing (positive case for LD-complexity reduction).** The single-SNP/member-count candidates failing while the reduced-test aggregated eMLG signal is the only residual is *expected*: outlier scans are prone to differentiation-driven false positives. The eMLG reduction is designed for higher power ($\approx 34\times$ fewer tests) and recovers coherent cluster-level signals invisible to the SNP scan (the eMLG-only class). The residual PC2 eMLG-only set ($\text{FDR} \approx 0.66$) is suggestive only.

6. Reproducibility (scripts & data objects)

Self-contained module folder `moduleB_BayPass/` with `R/` (scripts), `data/` (output objects), `Figures/` (plots) and `doc/` (this summary + `doc/tables/` candidate tables). Scripts are run **from the repo root** in the order below; they read shared/raw inputs from the repo root — the LD clustering `data/eMLG_5loci_0025_cM05.rds`, the genotype `data/hybrids*_maf005.Rdata`, and the BayPass run directories `with_aland/`, `aland_excluded/`, `aland_excluded_eMLG/` (which hold the per-run inputs, `run_baypass*.sh`, and the large `*_summary*.out`) — and write all module products under `moduleB_BayPass/`.

#	Step	Script (in <code>moduleB_BayPass/R/</code>)	Key output
1	BayPass inputs	<code>moduleB_prepare_{with_aland,aland_excluded,aland_excluded_eMLG}/</code> <code>$\rightarrow$ moduleB_write_baypass_inputs.R</code>	BayPass input files
—	BayPass runs (external)	<code>{with_aland,aland_excluded}/run_baypass.sh</code>	<code>Figures/manhattan_{DIEM*_summary*.out}</code>
2	SNP Manhattans	<code>moduleB_analyse_baypass.R</code>	<code>Figures/manhattan*_min5SigLoc*_png</code>
3	PC/ancestry confound	<code>moduleB_ancestry_confound.R</code>	<code>data/moduleC_ancestry_confound.{rds,png}</code>
4	Candidate definition + tables	<code>moduleB_climate_candidates.R</code>	<code>data/moduleB_climate_candidates.rds;</code> <code>doc/tables/climate_candidate_*.tex,pdf</code>
5	eMLG BayPass inputs	<code>moduleB_write_eMLG_baypass_inputs.R</code>	<code>aland_excluded_eMLG/u_eMLG.geno</code> (+ staged Ω /covariates)
—	eMLG BayPass runs (external)	<code>run_baypass_eMLG.sh</code>	<code>aland_excluded_eMLG/PC*_eMLG*_betai_reg.out</code>
6	eMLG Manhattan + outliers	<code>moduleB_eMLG_manhattan.R</code>	<code>data/moduleB_eMLG_association.rds;</code> <code>data/moduleB_eMLG_outliers.csv;</code> <code>Figures/moduleB_eMLG_manhattan.png</code>
7	Ω -structured sim-FDR (10k)	<code>moduleB_eMLG_null.R</code>	<code>data/moduleB_eMLG_null.rds</code>
8	FDR-overlay figures	re-run <code>moduleB_eMLG_manhattan.R</code> (reads step 7)	<code>Figures/moduleB_eMLG_manhattan.png</code> (triangles) + <code>Figures/moduleB_fdr_snp_manhattan.png</code>

Step 8: `moduleB_eMLG_manhattan.R` picks up `data/moduleB_eMLG_null.rds` if present, so re-running it after step 7 adds the FDR floor-survivor overlays (Figs 3–4). This summary PDF is rebuilt from `doc/` with `doc/render_moduleB_summary.sh`.

Reference for the LD-cluster / eMLG association approach: Li Z, Kemppainen P, Rastas P, Merilä J. 2018. Linkage disequilibrium clustering-based approach for association mapping with tightly linked genomewide data. *Mol Ecol Resour* 18:809–824.